MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 10 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. If the system of equations $3x + y = 1$ and $(2k-1)x + (k-1)y = 2k+1$ is inconsistent, then k equals: [1]

(A) -1 (B) 0 (C) 1 (D) 2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) 2** [0.5 Mark]

Justification: $\frac{3}{2k-1} = \frac{1}{k-1} \neq \frac{1}{2k+1} \implies 3k-3 = 2k-1 \implies \mathbf{k = 2}$ (with $1 \neq \frac{1}{5}$). [0.5 Mark]

Q2. The coordinates of the point where the line $2x - 3y = 12$ intersects the x-axis are: [1]

(A) $(0, -4)$ (B) $(6, 0)$ (C) $(-6, 0)$ (D) $(0, 4)$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B)** $(6, 0)$ [0.5 Mark]

Justification: Setting $y = 0$ (equation of x-axis): $2x - 0 = 12 \implies x = 6$, giving $\mathbf{(6, 0)}$. [0.5 Mark]

Q3. If $ax + 2y = 7$ and $3x + by = 18$ represent parallel lines, where a and b are integers, then ab equals: [1]

(A) 5 (B) 6 (C) 7 (D) 9

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 6** [0.5 Mark]

Justification: Parallel lines require $\frac{a}{3} = \frac{2}{b} \neq \frac{7}{18} \implies ab = 3 \times 2 = \mathbf{6}$ (since $a \in \mathbb{Z}$, $a/3 \neq 7/18$ always holds). [0.5 Mark]

Q4. A system of two linear equations in two variables is said to be dependent if it has: [1]

(A) no solution (B) exactly one solution (C) infinitely many solutions (D) exactly two solutions

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) infinitely many solutions** [0.5 Mark]

Justification: A dependent consistent system represents coincident lines and possesses **infinitely many solutions**. [0.5 Mark]

SECTION B: Assertion-Reason Questions — MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. **Assertion (A):** The solution of the pair of linear equations $x+y=3$ and $2x+5y=12$ is $x=1, y=2$.
Reason (R): A pair of values (x, y) that satisfies one equation of a system will automatically satisfy any other equation. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) (A) is true but (R) is false.** [0.5 Mark]

Verification: Checking $(1, 2)$: $1+2=3$ and $2(1)+5(2)=12$, so (A) is **true**. However, (R) is **false** (e.g., $(0, 3)$ satisfies $x+y=3$ but $2(0)+5(3)=15 \neq 12$). [0.5 Mark]

Q6. **Assertion (A):** If the perimeter of a rectangle is 30 cm and its length is 3 cm more than twice its breadth, then its breadth is 4 cm.
Reason (R): The algebraic formulation is $2(l+b)=30$ and $l=2b+3$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: $2(l+b)=30 \implies l+b=15$. Substituting $l=2b+3$: $(2b+3)+b=15 \implies 3b=12 \implies b=4$ cm. Both (A) and (R) are true and (R) explains (A). [0.5 Mark]

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. Solve $2x+3y=11$ and $2x-4y=-24$, and hence find the value of m for which $y=mx+3$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Solving for x and y [1 Mark]

Subtracting $2x-4y=-24$ from $2x+3y=11$: $7y=35 \implies y=5$. Then $2x+15=11 \implies 2x=-4 \implies x=-2$.

Step 2: Finding m [1 Mark]

Putting $(-2, 5)$ into $y=mx+3$: $5=m(-2)+3 \implies 2=-2m \implies m=-1$.

$x=-2$, $y=5$, $m=-1$

Q8. The larger of two supplementary angles exceeds the smaller by 18°. Find the angles. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Forming and Solving Equations [2 Marks]

Let larger angle $= x°$, smaller $= y°$. Since supplementary, $x+y=180°$, and $x-y=18°$.

Adding both: $2x=198° \implies x=99°$. Then $y=180°-99°=81°$. **99° and 81°**

Q9. A fraction becomes $1/3$ when 1 is subtracted from numerator and $1/4$ when 8 is added to denominator. Find the fraction. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Framing & Solving [2 Marks]

$\frac{x-1}{y}=\frac{1}{3} \implies 3x-y=3$ — (i) and $\frac{x}{y+8}=\frac{1}{4} \implies 4x-y=8$ — (ii).

Subtracting (i) from (ii): $x=5$. From (i): $3(5)-y=3 \implies y=12$. Fraction $= \mathbf{\frac{5}{12}}$

Q10. Solve for x and y: $152x - 378y = -74$ and $-378x + 152y = -604$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Adding & Subtracting Symmetric Equations [2 Marks]

Adding both equations: $-226x - 226y = -678 \implies x + y = \frac{-678}{-226} = 3$ — (i)

Subtracting Eq. 2 from Eq. 1: $530x - 530y = -74 - (-604) = 530 \implies x - y = 1$ — (ii)

Step 2: Solving Simple Pair [1 Mark]

Adding (i) and (ii): $2x = 4 \implies x = 2$. From (i): $y = 3 - 2 = 1$. $\boxed{\mathbf{x = 2, \quad y = 1}}$

Q11. Yash scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each incorrect answer, Yash would have scored 50 marks. How many questions were there in the test? [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Setting Up Score Equations [1.5 Marks]

Let right answers $= x$ and wrong answers $= y$.

$3x - y = 40$ — (i) and $4x - 2y = 50 \implies 2x - y = 25$ — (ii)

Step 2: Solving & Finding Total Questions [1.5 Marks]

Subtracting (ii) from (i): $x = 40 - 25 = 15$. From (i): $3(15) - y = 40 \implies y = 5$.

Total questions attempted $= x + y = 15 + 5 = \boxed{\mathbf{20}\text{ questions}}$.

Q12. A man sold a chair and a table together for Rs. 1520 making a profit of 25% on the chair and 10% on the table. By selling them together for Rs. 1535 he would have made a profit of 10% on the chair and 25% on the table. Find the cost price of each. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let CP of chair = Rs. x and CP of table = Rs. y.

Formula / Theorem Used: Selling Price (SP) $= \text{CP} \times \left(1 + \frac{\text{Profit \%}}{100}\right)$

Step 1: Forming Selling-Price Equations [1.5 Marks]

$1.25x + 1.10y = 1520 \implies 125x + 110y = 152000$ — (i)

$1.10x + 1.25y = 1535 \implies 110x + 125y = 153500$ — (ii)

Step 2: Symmetric Add-Subtract Solution [1.5 Marks]

Adding (i) and (ii): $235(x + y) = 305500 \implies x + y = 1300$ — (iii)

Subtracting (i) from (ii): $15(y - x) = 1500 \implies y - x = 100$ — (iv)

Adding (iii) and (iv): $2y = 1400 \implies y = 700$, $x = 1300 - 700 = 600$.

$\boxed{\text{Chair CP} = \text{Rs. } \mathbf{600}, \qquad \text{Table CP} = \text{Rs. } \mathbf{700}}$

Q13. A part of monthly hostel charges in a college is fixed and the remaining depends on the number of days one has taken food in the mess. When student A takes food for 20 days, he has to pay Rs. 1000 as hostel charges whereas student B, who takes food for 26 days, pays Rs. 1180. Find the fixed charges and the cost of food per day. If student C takes food for 28 days, find the total charges he has to pay. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let fixed monthly charge = Rs. x and food charge per day = Rs. y.

Step 1: Setting Up Equations [2 Marks]

Student A: $x + 20y = 1000$ — (i) Student B: $x + 26y = 1180$ — (ii)

Step 2: Solving for y and x [2 Marks]

Subtracting (i) from (ii): $6y = 180 \implies y =$ Rs. 30/day.

Putting $y = 30$ in (i): $x + 20(30) = 1000 \implies x = 1000 - 600 =$ Rs. 400.

Step 3: Total Charges for Student C (28 Days) [1 Mark]

C's total $= x + 28y = 400 + 28(30) = 400 + 840 =$ Rs. 1,240.

Fixed = Rs. **400**, Food/day = Rs. **30**, C's bill = Rs. **1,240**

Q14. CASE STUDY: Competitive Examination Evaluation & Negative Marking

In an Olympiad scholarship examination of 60 questions, candidate marks are calculated under a differential scoring scheme. A student receives 4 marks for every correct answer, while 1 mark is deducted for every incorrect response. Kabir attempted all 60 questions and scored a net total of 140 marks. Let x denote the number of correct answers and y denote the number of incorrect answers. [4]

(i) Write the system of linear equations in x and y representing the total questions attempted and Kabir's score. [1]

(ii) Determine how many questions Kabir answered correctly (x). [1]

(iii) Determine how many questions Kabir answered incorrectly (y). [1]

(iv) If negative marking is increased to 2 marks per wrong answer, what would Kabir's score be for the same correct answers? [1]

— OR —

Find the minimum number of correct answers needed to score at least 180 marks with 1 mark penalty. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): System of Linear Equations [1 Mark]

Total questions attempted: $x + y = 60$ — (i) Net score: $4x - y = 140$ — (ii)

Sub-part (ii): Correct Answers (x) [1 Mark]

Adding (i) and (ii): $5x = 200 \implies$ $\boxed{\mathbf{x = 40}\text{ correct answers}}$.

Sub-part (iii): Incorrect Answers (y) [1 Mark]

From (i): $y = 60 - 40 =$ $\boxed{\mathbf{y = 20}\text{ incorrect answers}}$.

Sub-part (iv): Score Under 2-Mark Penalty [1 Mark]

New Score $= 4x - 2y = 4(40) - 2(20) = 160 - 40 =$ $\boxed{\mathbf{120}\text{ marks}}$.

Sub-part (iv) [OR Option]: Minimum Correct Answers for ≥ 180 Marks [1 Mark]

Under all 60 attempted ($y = 60 - x$): $4x - (60 - x) \geq 180 \implies 5x \geq 240 \implies x \geq 48$.

Check at $x = 48, y = 12$: $4(48) - 12 = 180$. Hence minimum needed = $\boxed{\mathbf{48}\text{ correct answers}}$.

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